Roll No. | | | | | | | |

ODD END SEMESTER EXAMINATION 2024

Name of the Course: B.Tech

Semester: 7th

Name of the Paper: Data Warehousing and Data Mining

Paper Code: TCS -722

Time: 3 Hours

Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub-questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20marks)	**CO1**
(a)	Define a Data Warehouse. How is it different from a database system?	
(b)	Describe the Multi-Dimensional Data Model with examples.	
(c)	Differentiate between Star Schema, Snowflake Schema, and Fact Constellations.	
Q2	(20 marks)	**CO2**
(a)	Compare and contrast ROLAP, MOLAP, and HOLAP servers.	
(b)	Explain the importance of Backup and Recovery in Data Warehousing.	
(c)	Explain the purpose of Aggregation in Data Warehousing with examples.	
Q3	(20 marks)	**CO3**
(a)	What is Data Preprocessing? Why is it important in Data Mining?	
(b)	Explain the process of Data Generalization with an example.	
(c)	What are the common techniques for dealing with noisy data?	
Q4	(20 marks)	**CO4**
(a)	What is the Apriori Algorithm? Explain it with a detailed example.	
(b)	Describe the working of a Decision Tree algorithm with an example.	
(c)	Define Data Mining and its functionalities.	
Q5	(20 marks)	

(a)	Explain how supervised and unsupervised classification methods differ in terms of data requirements, approach, and outcomes. Provide examples to illustrate their distinct applications.	CO5
(b)	Consider a healthcare organization aiming to predict disease risks based on patient records. Discuss how machine learning techniques can be utilized to improve both the accuracy of classification (e.g., disease diagnosis) and prediction (e.g., risk levels).	
(c)	You are analyzing a financial dataset for building a credit risk model, but several entries have missing values in critical fields like income and credit history. Describe the methods you would use to handle these missing values.	